

JKBOSE 2026

CLASS 11TH

RESPIRATION IN PLANTS

LECTURE-02

BOTANY

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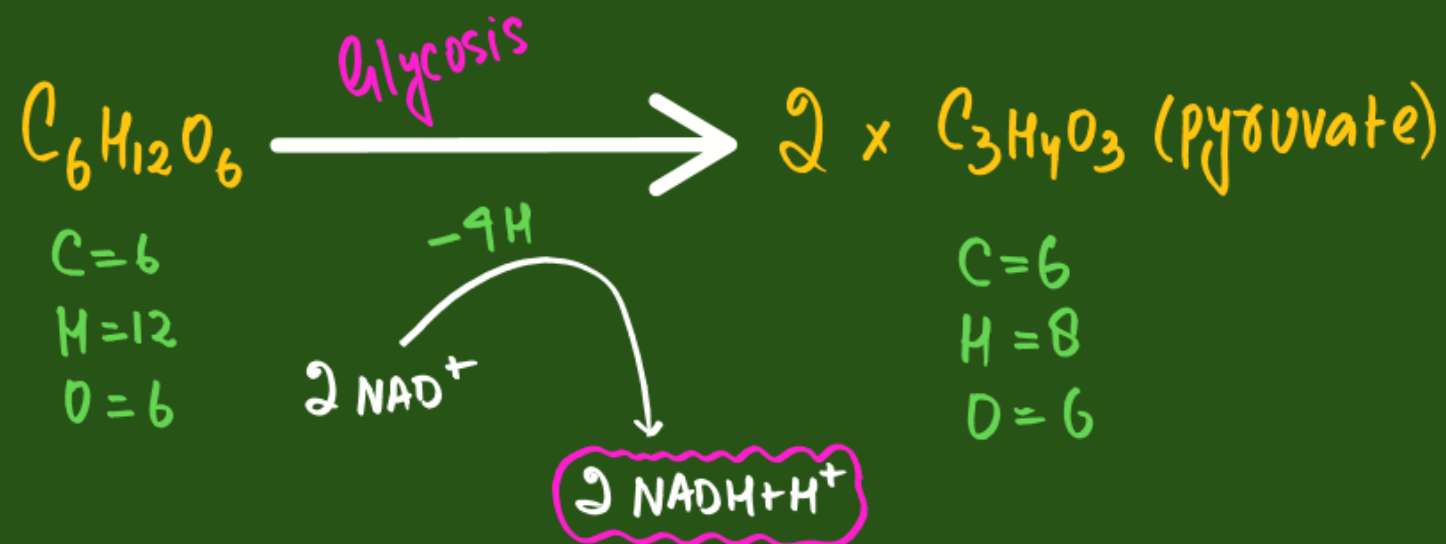


Topics To Covered

- 1** Glycolysis
- 2** Fate Of Pyruvate
- 3** Fermentation

Glycolysis

Glyco's = Glucose
lysis = Splitting



≡ Glycolysis involves partial/incomplete oxidation of glucose to form 02 molecules of pyruvate.

≡ During glycolysis "4H" atoms are lost which are taken by 2 molecules of NAD^+ to form $2(\text{NADH}+\text{H}^+)$

≡ Site of glycolysis = Cytoplasm of prokaryotic / eukaryotic cell

≡ Glycolysis is universal process i.e., it occurs in all prokaryotes and eukaryotes.

≡ It is common to both aerobic and anaerobic respiration.

≡ Glycolysis is a 10 stepped rxn. (Enzymatically Controlled).

10 Reactions of Glycolysis

Two Phases

Preparatory phase (rxn 01-05)

- ATP is used.

Pay-off phase (rxn 6-10).

- ATP is formed.

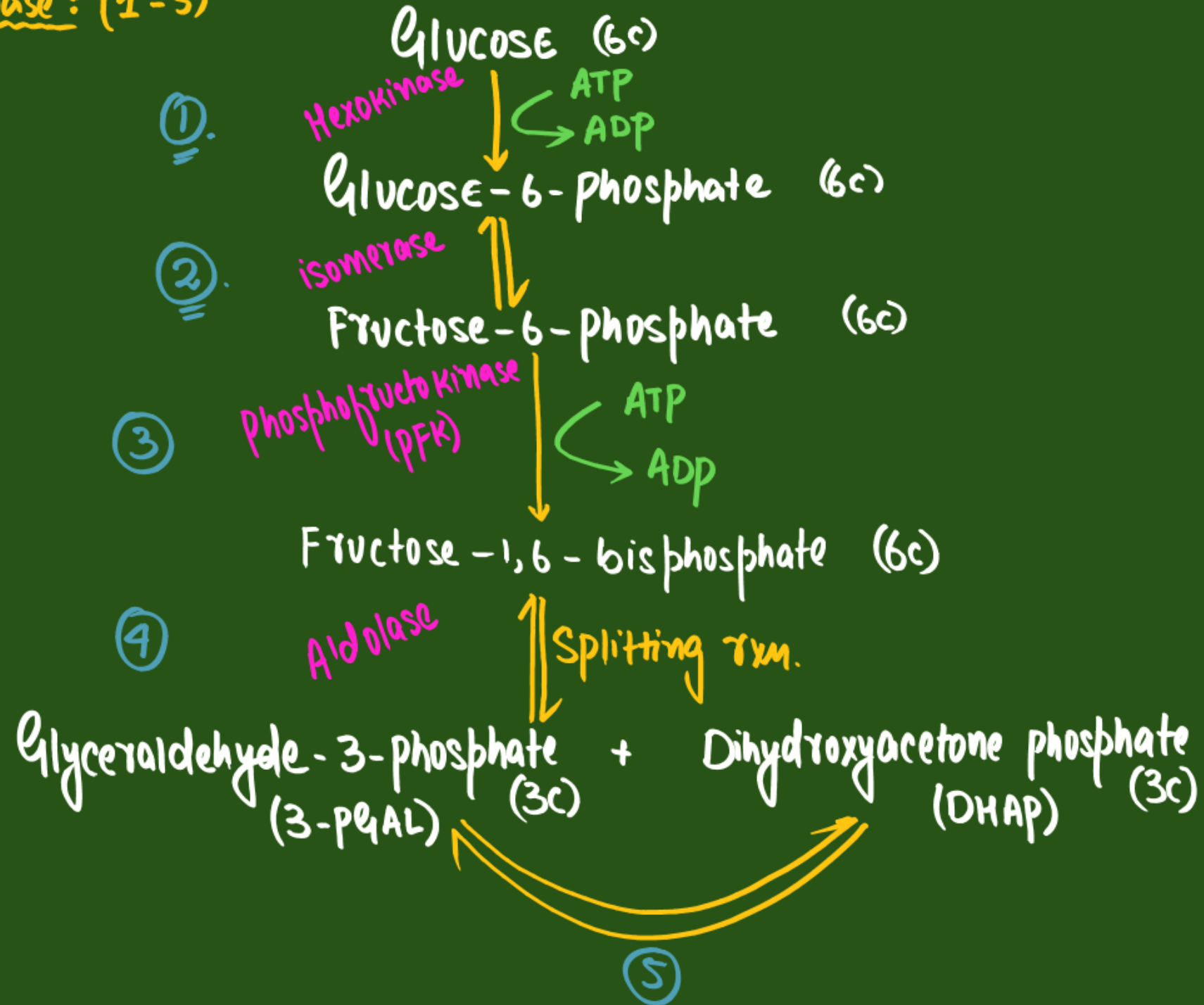
≡ Metabolic pathway of Glycolysis was worked out by 03 Scientists:

- Embden
- Meyerhof
- Paras

i.e, why Glycolysis is also called as "EMP" pathway.

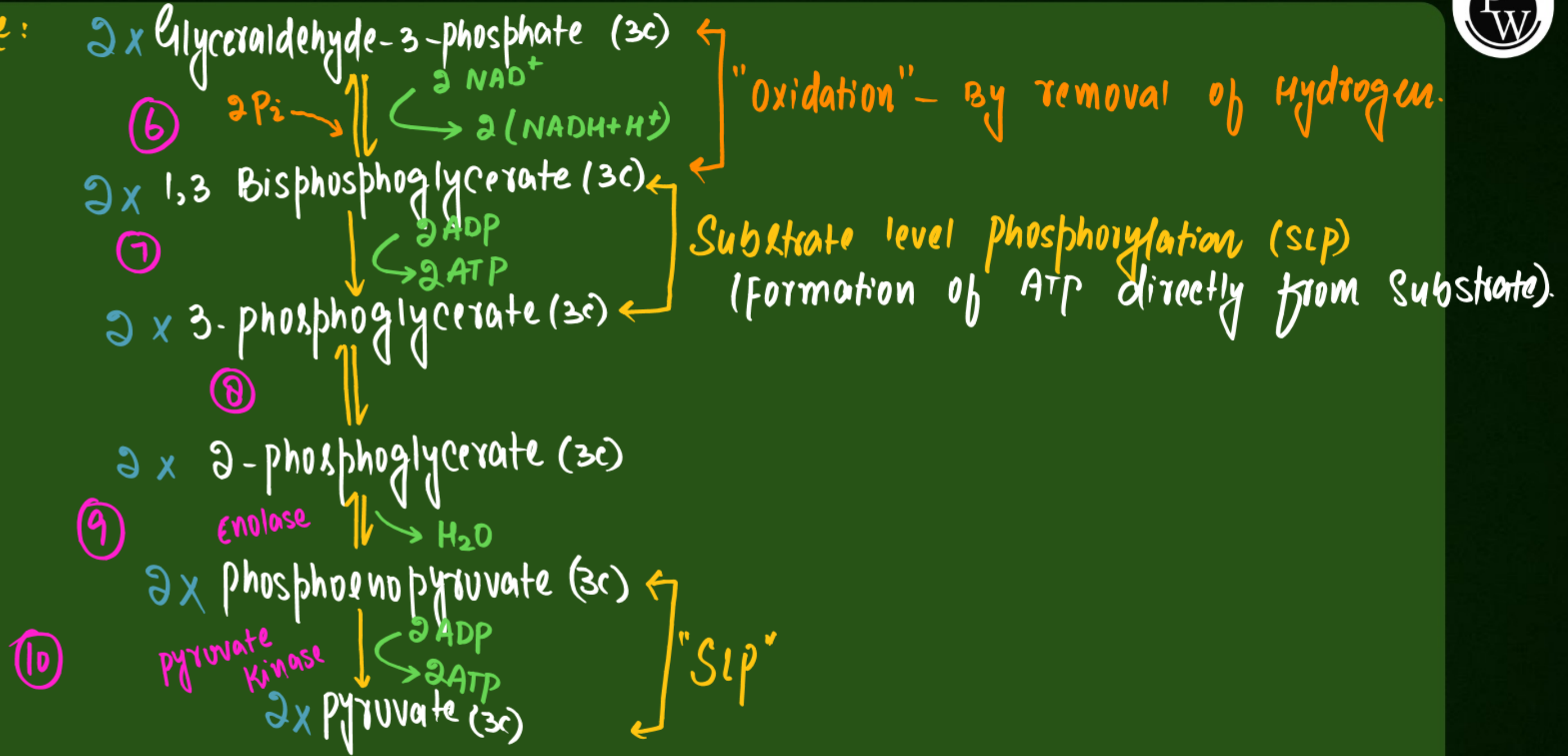
Metabolic Pathway of Glycolysis

A) Preparatory Phase: (1-5)



Note: At the end of 5th rxn. we have two mol. of 3-PGAL.

B. Pay off Phase:
(6-10)



During Glycolysis

• 01 molecule of glucose gives :

a. 2 x pyruvate ($C_3H_4O_3$)

b. NO CO_2 .

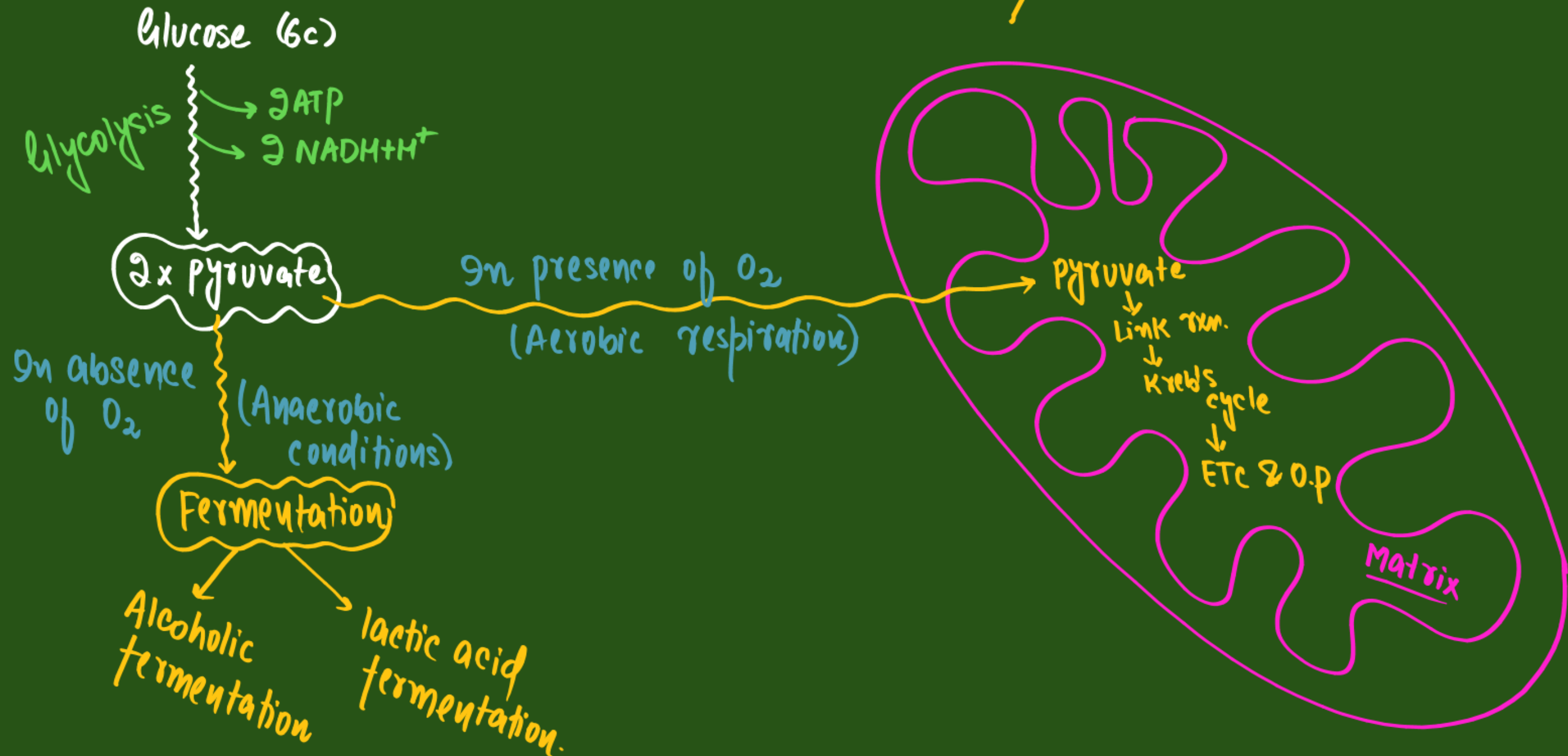
c. 2 ($NADH + H^+$) $\longrightarrow$ "at rxn. 6"

$\therefore$ ATP used = 02 (at rxn. ① and rxn. ③)

$\therefore$ ATP generated = 04 (at rxn. ⑦ and rxn. ⑩)

$\therefore$ Net gain of ATP during glycolysis = $4 - 2 = 02$ ATP.

Fate Of Pyruvate



- Pyruvic acid is key product of glycolysis.
- Metabolic fate of pyruvate depends on cellular need.
- Different cells handle pyruvic acid by 03 ways:

a. lactic acid fermentation

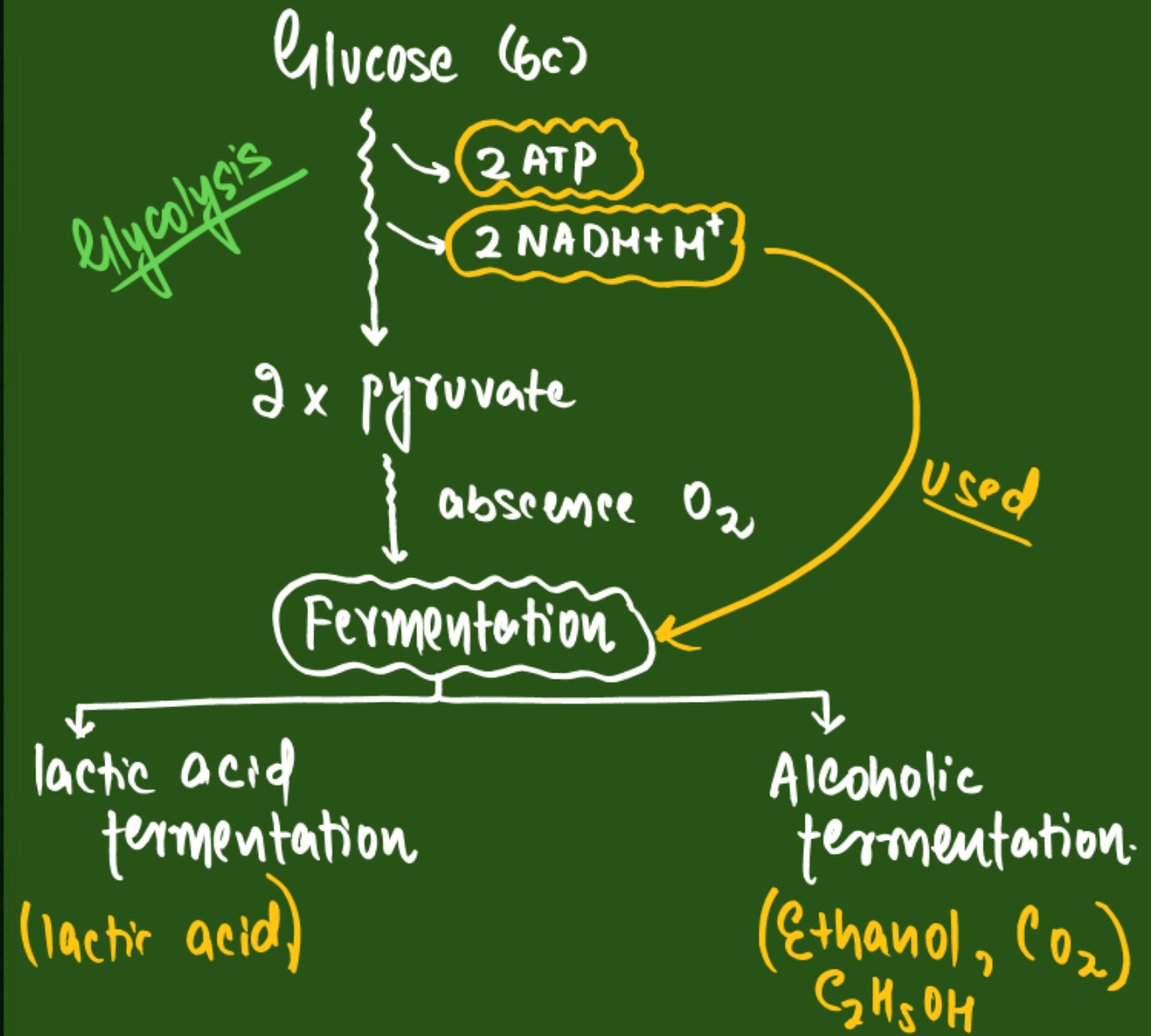
b. Alcoholic fermentation.

c. Aerobic respiration.

} Under anaerobic conditions in many prokaryotes and unicellular eukaryotes (yeast).

- lactic acid fermentation also takes place in muscle cells where mitochondria is present but O_2 is in less concentration (during exercise).

Fermentation



≡ In fermentation, incomplete oxidation of glucose takes place by a set of reactions which results in formation of compounds like - lactic acid, ethanol, CO₂.

≡ Net gain of ATP during fermentation = 2 ATP

A. Lactic Acid Fermentation:

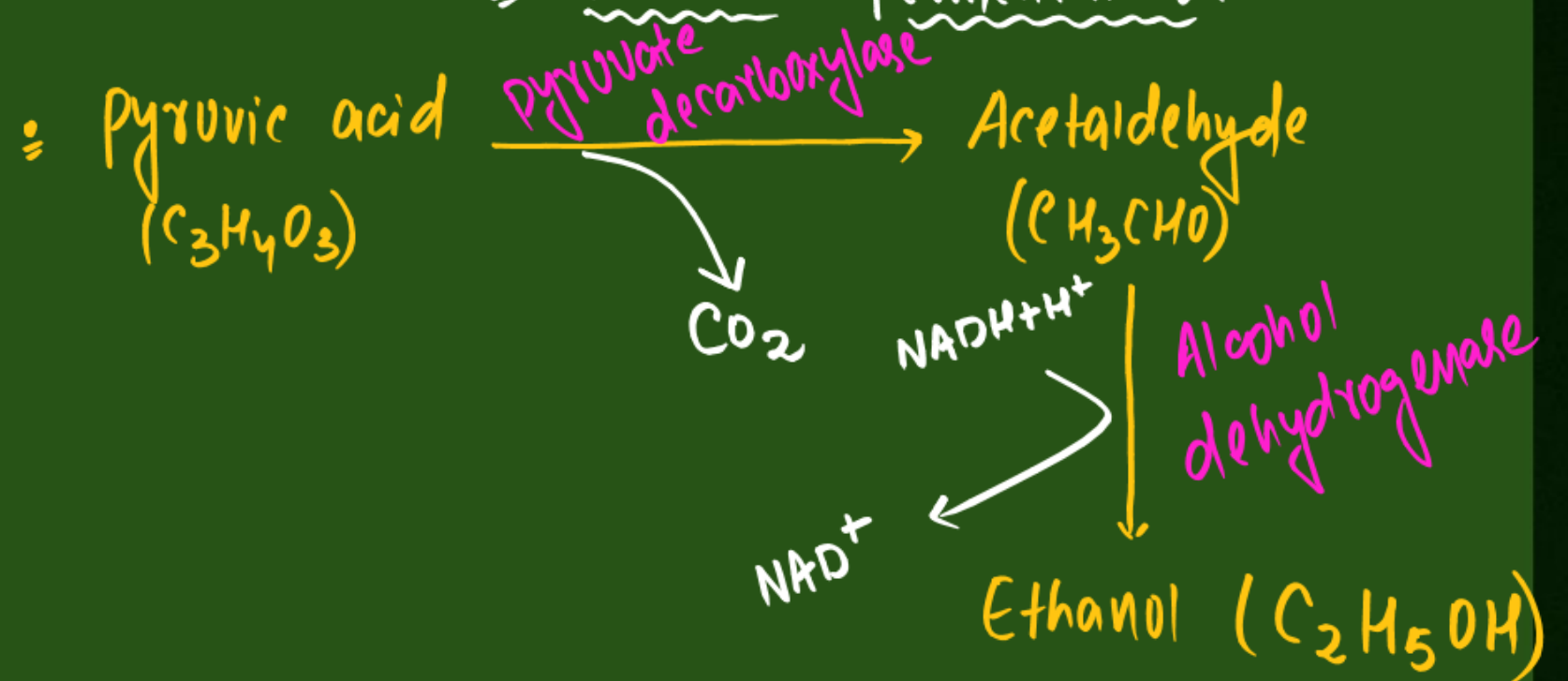


• Final e^- acceptor is pyruvic acid.

• Ex. Muscle cells, Bacteria.

• Site: Cytoplasm.

B. Alcoholic Fermentation:



• Final e^- acceptor is Acetaldehyde.

• Ex. Yeast.

• Site: Cytoplasm.

≡ Net gain of ATP during fermentation = 2 ATP

≡ In both fermentations, less than 7% of energy is released.

$$\frac{2 \text{ ATP}}{38 \text{ ATP}} \times 100 = 5.26\% \text{ which is less than } 7\%.$$

≡ Yeast poison themselves to death, when concentration of ethanol reaches about 13% in their cell/body. That is why naturally occurring alcoholic beverages contain 13% ethanol.

- If greater concentration (>13%) of ethanol is required, then **distillation** of naturally occurring beverage is done.

QUESTIONS

Assertion (A): ATP is used at two steps in glycolysis. ✓ **[MR*] (2023)**

Reason (R): First ATP is used in converting glucose into glucose-6-phosphate and second ATP is used in conversion of fructose-6-phosphate into fructose-1-6-bis phosphate.

- A** A is false but R is true.
- B** Both A and R are true and R is the correct explanation of A
- C** Both A and R are true but R is NOT the correct explanation of A.
- D** A is true but R is false.

QUESTIONS

What amount of energy is released from glucose during lactic acid fermentation? (2022)

- A** Less than 7%
- B** More than 18%
- C** Approximately 15%
- D** About 10%

$$\frac{2 \text{ ATP}}{38 \text{ ATP}} \times 100$$

$$= 5.26\% < 7\%$$

HOMEWORK



Revision

Thank
You